

(b)

Obstacles to both chaining and ranging

5. Measure DE. That is the obstructed distance between BG. (DE=BG)

3.2.7 Errors in chaining

The common types of errors are:

1. Mistakes
2. Compensating errors (or) accidental errors (or) random errors
3. Cumulative errors (or) systematic errors

1. Mistakes

Mistakes occur due to displacement of arrow, miscounting the number of chains, misreading the chain length, erroneous recording, etc.

2. Compensating errors

They are liable to occur in either direction and hence tend to compensate. They do not affect results much.

These errors are caused by the following reasons

i. Incorrect holding of the chain
ii. If the fractional part of the chain may not be correct
iii. Tape is not calibrated uniformly throughout its length.

3. Cumulative errors

These are the errors which are liable to occur in the same direction and tend to accumulate. Cumulative errors occurred due to

1. Bends in links, knots in links, removal of rings during adjustment of chain.
2. Slope correction is not applied when measuring along sloping ground.
3. Sag correction is not applied when chain is suspended in air
4. Measurements are made along the incorrectly aligned line.

3.3 Levelling

3.3.1 Definition

Levelling is the art of determining the relative vertical heights of different points on the surface of earth. Hence, in levelling, the measurements are taken only in the vertical plane.

3.3.2 Basic terms used in levelling

1. Level surface

It is defined as curved surface which is parallel to the mean spheroidal surface of the earth.

2. Level line

It is a line lying in the level surface. All points on the level line are at the same elevation.

3. Horizontal line

It is a straight line lying in the horizontal plane. It is tangential to the level line and perpendicular to the vertical line at that point.

4. Horizontal plane

It is a plane perpendicular to the direction of gravity. It is tangential to the level surface at that point.

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